

Investigating the efficacy of Artificial Intelligence

**Is Artificial Intelligence essential to
the future of Cybersecurity and
Threat Detection?**

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Abstract

Aims of the Dissertation

This dissertation investigates whether the recent developments in Artificial Intelligence are sufficient to justify its incorporation into modern Cybersecurity techniques. This dissertation explores the importance of Cybersecurity and current measures that are used by companies. Following this, I aim to investigate how Artificial Intelligence can be integrated into current Cybersecurity measures and then discuss of the opinions for and against their implementation. My key question is to decide whether to implement or prevent using Artificial Intelligence in Cybersecurity measures.

Why is this Important

While Artificial Intelligence is not a new idea, initially being coined by a 1955 'Artificial Intelligence' workshop held by John McCarthy at Dartmouth College, the term has recently gained significant popularity, with key examples being the AlphaGo vs Le Sedol match in the strategy board game Go, which saw the Google DeepMind developed computer being victorious in four of the five games against the 9-dan professional Go player. In less than a decade since this Science Breakthrough of the Year, large language models like OpenAI's ChatGPT and Microsoft's Copilot have swiftly gained popularity and investment, which has led to new job titles relating to Artificial Intelligence, such as Machine Learning Engineers and Artificial Intelligence research scientists. This has opened discussions into how Artificial Intelligence can be incorporated into various sectors of the economy and our society, which is evidenced by research from the joint efforts of the UK Government Department for Digital, Culture Media & Sport and the Office of Artificial Intelligence which became a branch of Department for Science, Innovation and Technology which found 15% of all businesses, or 432,000 firms, adopting at least one Artificial Intelligence technology.

Another important sector, Cybersecurity, derives from a need to protect computer software, systems, and networks against malicious attackers. This makes it integral to every organisation's operation, and it has led to discussions on how to improve Cybersecurity methodologies to protect against the vastly growing threats against such systems. As a result, this has sparked debates about combining the two important sectors of Cybersecurity and Artificial Intelligence to promote new and more effective Cybersecurity methodologies.

Introduction and Rationale

Artificial Intelligence is a rapidly developing area of Computer Science that has existed, at least in concept, for decades. It focuses on creating computer systems or models that can complete tasks that would usually require highly skilled human intelligence. This differs from traditional computer systems, which are typically efficient at repetitive tasks that do not change frequently, such as calculations and running applications. Cybersecurity is another extremely important sector of Computer Science. It involves using systems to protect private individuals and organisations from cybercriminal attacks which aim to access, change, or destroy sensitive information. The Cybersecurity industry is incredibly valuable, at an estimated USD \$222.66 billion in 2023 and is estimated to grow at a compound annual growth rate of 12.3% from 2023 to 2030, according to Grandview Research (2023), proving the value placed on such measures.

Real-time threat detection is an important preventative measure against cyber-attacks, aiming to stop the attack before it can take place. Real-time monitoring is also essential for malware detection and can be a significant challenge to implement due to its requirement for significant resources and accuracy issues which result in false positives (where normal activity is incorrectly flagged as malicious, wasting time and resources or preventing normal programs from working properly) and false negatives (where malicious threats are undetected leading to risks in the system). A key negative of current Cybersecurity measures such as anti-malware programs is that they rely on its private database of malware, which makes them vulnerable to attacks that are slightly altered or haven't been seen before, called "Zero-Day" attacks, meaning Cybersecurity professionals must constantly work to 'catch up' with cybercriminals to fight against their exploits, which is a key area in which Artificial Intelligence is expected to make a difference. Furthermore, there is an increasing risk of cybercriminals using Artificial Intelligence models to enhance their attacks.

Literature Review

Foundational Concepts

Cybersecurity, as defined by the National Institute of Standards and Technology (NIST), refers to the practice of safe methods of protecting and using systems to protect yourself from cyber-attacks – usually aimed at accessing, modifying or destroying sensitive information, whether that be for money or interrupting business procedures (Barrett, 2018). These methods involve a wide variety of technologies, practices and standards designed to protect an organisation or individual's digital assets from unauthorised access.

Artificial Intelligence can be broken down into various subfields, such as Machine Learning and Natural Learning Processing (Russell & Norvig, 2016). As highlighted, Machine Learning is a key subset of Artificial Intelligence and is key to the recent evolution of Cybersecurity methods. It involves training algorithms on base data to make decisions without being explicitly programmed to do so like a functional program would (Mitchell, 1997). Natural Language Processing is also a key subset of Artificial Intelligence. It is involved with programs which can understand and communicate with human language, such as the various Open AI "ChatGPT" models. It is also important to note that Artificial Intelligence models use a probabilistic approach where they calculate the "answer" with the greatest probability of being correct based on its previous training rather than "knowing" the answer.

However, it is important to consider the conceptual biases of these sources. As outlined above, Artificial Intelligence is not a new topic or concept, thus, meaning the linked texts may outline theories or definitions of Artificial Intelligence and Cybersecurity concepts. This is evident in how Cybersecurity challenges are constantly changing with new challenges constantly developing. Furthermore, there has been discussion into how to define "intelligence" itself which could, in turn, mean that their definition of Artificial Intelligence is inaccurate purely

based on your definition of “intelligence,” which varies wildly with your geolocation and culture, particularly with whether “intelligence” is the ability to be able to problem solve, a feature which could be argued traditional programs also have, or includes the ability of creativity and emotional knowledge. Depending on these definitions, this could lead to different ideas on how Artificial Intelligence can be used, and different methodologies for Cybersecurity could be overlooked. However, it is also important to note these texts are foundational texts in Artificial Intelligence and machine learning and have also been peer-reviewed, making them highly reliable.

Applications of Artificial Intelligence in Cybersecurity and Threat Detection

A primary area where Artificial Intelligence has been used to evolve Cybersecurity is threat detection. This is attributed to the advancement of Machine Learning algorithms, a subset of Artificial Intelligence, particularly in the identification of anomalies and potential threats, a key first step in Cybersecurity procedures. This is shown by Buczak and Guven (2016), who state in their survey that Machine Learning can analyse a large amount of data to detect patterns of cyber security threats that traditional methods would otherwise miss. This is particularly helpful in Zero-Day attacks - an attack where attackers exploit a flaw in their program or systems before developers have a chance to address it – allowing these attacks to be prevented before causing significant damage, without the need to use Business Continuity Plans - defined by Mary K. Pratt et al (2023) as a playbook designed to help an organisation quickly resume business functions during the face of disruption. This aims to keep organisations running smoothly, maximising their productivity and productive potential by reducing downtime caused by malicious cyber-attacks.

Ahmad, Hadgkiss, and Ruighaver (2012) discuss how AI models can set a baseline for normal network behaviour and detect changes that may identify a malicious intruder. This is proven by Artificial Intelligence’s clear ability to adapt and learn, making it suitable for anomaly

identification. Such models can learn and adapt continuously, particularly for data they gain over time, increasing their detection accuracy. This is furthered by (Buczak & Guven, 2016), who suggests unsupervised learning methods – a type of machine learning which learns from data without requiring human supervision- allowing Artificial Intelligence models to find new types of threats without prior knowledge of their existence.

Sommer and Paxson (2010) highlight using Artificial Intelligence to prevent and predict future attacks by analysing past attacks to find patterns that suggest imminent threats to an organisation's sensitive data and information. This can be an essential improvement to a firm's Cybersecurity ideals as it allows them to proactively fortify their defences in anticipation of breaches, minimising the damage caused in the case of an attack. This type of analysis can be described as a predictive analysis done by leveraging historical data to generate patterns that would indicate threats.

Another use of Artificial Intelligence in the evolution of Cybersecurity is in Endpoint Security. Endpoint security involves protecting devices (endpoints) that act at the end of a network, such as workstations, servers, or laptops. Shashank and Purnima (2019) claim Artificial Intelligence endpoint protection platforms can continuously monitor device security well as detect threats and autonomously respond to threats. This allows particularly vulnerable endpoints, such as laptops and desktop computers, to be constantly protected against malware and attacks. Shrobe et al. (2018) emphasise that Artificial Intelligence systems can be used to react to threats in real-time, reducing the time that attackers have to conduct their attacks. They may also be used to conduct preliminary forensic analysis to understand the nature of attacks.

Another key subset of Artificial Intelligence is Natural Language Processing. As described by Mittal et al. (2016), Natural Language Processing is a branch of Artificial Intelligence which is used to enhance threat perception by analysing unstructured data, which allows the Artificial Intelligence model to learn from a wider range and type of media sources. They describe how

such techniques can extract relevant information about emerging threats, enhance the intelligence of an organisation or third-party security team, allow them to advance ahead of malicious actors, and allow them to understand and protect against new threat tactics.

While these applications and sources in this section help to provide up-to-date data on the application of Artificial Intelligence in Cybersecurity and are published in peer-reviewed journals and published by highly respected organisations such as the Institute of Electrical and Electronics Engineers. There may have been bias in the algorithms or data published in their studies, such as Buczak & Guven, 2016 and Summer & Paxson, 2010 which may have underreported false positives or negatives from biased training data. Furthermore, studies like these typically are subject to publication biases, the idea that positive results are more likely to be published, which may lead to an overly optimistic view of the capabilities of Artificial Intelligence models in the world of Cybersecurity.

Challenges and Limitations of Artificial Intelligence in Cybersecurity

On the other hand, there are key limitations and challenges that stem from the integration of Artificial Intelligence in Cybersecurity. A significant concern is that malicious actors may attack the Artificial Intelligence systems themselves. Szegedy et al. (2014) demonstrate how malicious actors can use exploitations and vulnerabilities in Artificial Intelligence models, spoiling them with false information, causing them to misidentify threats, and allowing for easier evasion of detection. In addition to this, the success of such Artificial Intelligence models heavily depends on the quality of the information provided along with the quantity, meaning there is a threat for malicious actors to spoof the Artificial Intelligence model's training data, especially with unsupervised training models. These can result in false positives and, overall, inaccurate models. This has meant there has been a large demand for tamper-proof Artificial Intelligence models.

Conclusion

Artificial Intelligence in Cybersecurity has become an increasingly practical method for keeping sensitive data safe and maintaining its integrity. There are several ways to approach the topic of Artificial Intelligence, particularly in its subsets of Machine Learning and Natural Language Processing. However, there are some important challenges that must be overcome before Artificial Intelligence can become even more useful in the evolution of Cybersecurity. It is also important to note the biases which may stem from different sources, such as the conceptual bias of early works, which may have out-of-date concepts, as well as a contextual bias, which may promote better narratives of the abilities of Artificial Intelligence and so may not represent all evidence.

Why is Cybersecurity so essential, and what are the effects of a lack of Cybersecurity?

Cybersecurity is an important aspect of business planning for every business as they aim to protect the confidential information of the business, such as financial records, marketing strategies and information of their customers and workers according to the specifications of the CIA Triad (Van der Ham, 2021). This describes the three main attributes of secure systems: Confidentiality – where sensitive data is protected so unauthorised users cannot access it, Integrity- ensuring that data is kept accurate and ensuring different ensuring that if there are any changes, they can be linked back to an individual or organisation, and Availability- Making sure that data is available to authorised users whenever they want to access it. If these attributes are not present in a company's Cybersecurity measures, the firm may see confidential information in unauthorised hands. Such information can be used for a variety of reasons, such as the sale of confidential business secrets to competitors of the business. This could result in a fall in their competitive advantages as competing businesses adopt productive methods and reduce their costs, so they are able to compete better. This can mean the original firm may see financial losses as they lose their ability to compete. As well as this, with the sale of private personal data through a data breach, the firm may see significant reputational damage leading to a loss of customer loyalty, which can severely impact the ability of the business to survive in the long run.

Businesses may also face legal action or fines if their Cybersecurity measures are not sufficient and there is a breach of personal information. A key example of this was after the 2011 Sony PlayStation Network outage, colloquially known as the 'PSN hack,' where an external extrusion led to the PlayStation gaming network going offline for over three weeks, preventing millions of users from playing their video games online. In addition to this, the BBC (2013) reported that it

resulted in over 77 million accounts and their private information being compromised. As a result, Sony was fined £250,000 by the UK Government for breaches of the 1998 Data Protection Act. An attack as successful as the PSN hack is also likely to cause large reductions in the confidence of their customers, which may, in turn, result in customers switching platforms to one of their competitors and seeing reductions in new sign-ups and purchases.

Furthermore, malicious actors may also attempt to cause significant damage to internal systems or networks. One of the main examples of this was the significant May 2017 WannaCry ransomware attack. This attack exploited insecurities in devices which used older and vulnerable Microsoft Windows operating systems. This attack was estimated by the European Union Agency for Law Enforcement Cooperation (2017), also known as Europol, to have affected 200,000 computers across 150 countries and caused major disruptions to the services of businesses across the globe. A key example was the Nissan Motor Manufacturing UK factory in Tyne and Wear, which had to completely halt production after the ransomware infected its systems. However, one of the largest agencies struck by this attack was the National Health Service in the UK, where computers, MRI scanners and blood-storage refrigerators were affected. This meant that non-critical emergencies were turned away and ambulances were diverted, evidencing the true danger of Cybersecurity attacks as lives were affected and long NHS waiting times were already worsening.

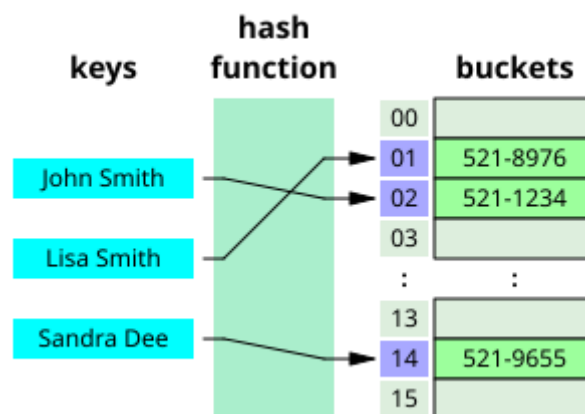
On an individual level, a lack of personal Cybersecurity methods can also result in several severe risks, such as a loss of personal information, or the information of your contacts, through the exploitation of phishing and social engineering attacks. Such attacks aim to gain unauthorised access to accounts or information through the exploitation of human vulnerabilities, like convincing a victim that they are someone they are not, including their bank or a loved one, to get the victim to click a link or leak confidential information like passwords or contact information. Such information can then be used to spread the attack onto others,

including people in their contacts, or then to commit further attacks, including identity theft.

Other types of attacks may aim to brute-force into a specific individual's account. As a result, this advocates for the importance of Cybersecurity methods on a personal level, for example training in identifying and dealing with phishing and social engineering attacks, as well as safe and secure passwords and online habits to prevent accounts from being breached by brute-force attacks and reducing the effect of data breaches, as described in the next topic.

How do current Cybersecurity Measures work?

Current Cybersecurity measures aim to mitigate the risks of attacks and ensure data integrity using a variety of different software and physical protection methods. One key software protection method is anti-malware programs such as Malwarebytes and Norton Anti-Malware. Anti-malware programs are built from a large database of hashes which uniquely identify a specific malware. Hashing is the process by which your data or text is passed through a hashing function, creating a characterless text which cannot be decoded and is then placed into a hash table or database for storage. but. The process of this is shown in the figure (Stolfi, 2009) which shows names of contacts being passed through the hash function to be assigned unique indexes in the hash table. This allows for easy comparison as ideally each hash should be unique and should change vastly even if the initial keys are similar.

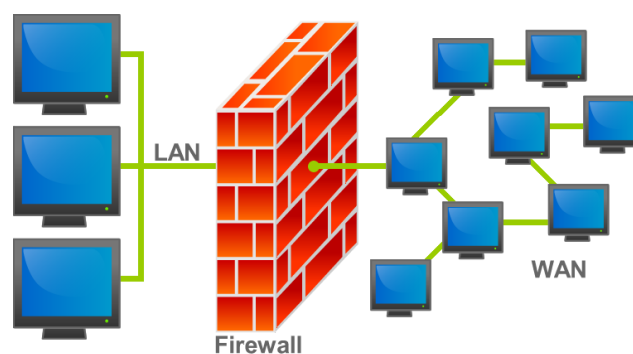


A hash table of contacts (Stolfi, 2009)

In Anti-Malware systems, program files are then hashed and compared to the hashes in the database to check whether snippets of the code are the same to the unique virus or behave in a similar nature. The software then flags and quarantines suspicious files for the user's review and removal. A benefit of this is that the hash database is a one-way function meaning the original malware cannot be reengineered and used for malicious reasons. However, because anti-malware software relies on its own private database, it is only effective if it is up to date, which means that newer viruses will not be protected against because they are not available on

the database. Anti-malware systems can also consume a significant amount of system resources, such as memory, meaning that it can take a significant amount of time for system scans to take place and can cause the whole system to slow down, reducing the ability of users to do work on their computers. There is also a significant risk of false positives where legitimate files are flagged as malicious and, therefore, are quarantined and removed from a user's computer. This can cause a significant challenge and disruption to an organisation or individual user's work.

In addition to this, firewalls contain software which aims to control traffic entering a network. It does this by scanning network packets, where data has been broken up into small packets to then be transferred across the internet. This is shown in the illustration (Pedrozo, 2007) which shows how the firewall acts as a barrier or "wall" between the local network (LAN) and wide-area network (WAN) such as the internet.



An illustration of a firewall (Pedrozo, 2007)

Previously set security rules then prevent certain packets from entering the internal network to ensure that only safe traffic is allowed into the firewall. However, like anti-malware software, a key negative of firewalls is false positives, which negatively impact the usage of the computer as legitimate traffic is blocked, reducing a user's productivity if the set security rules are not specific enough, which can be hard to implement for a large organisation which has a large and diverse amount of job roles which a variety of different tasks.

Physical protection methods revolve around three main authentication methods: what you are, what you have, and what you know. At least two different methods are used to allow authorised users into a system or protected area of a building, called two-factor authentication. Examples of what you include biometric authentication methods such as iris scanning, or fingerprint scanning and so requires a significant amount of specialised equipment to enforce.

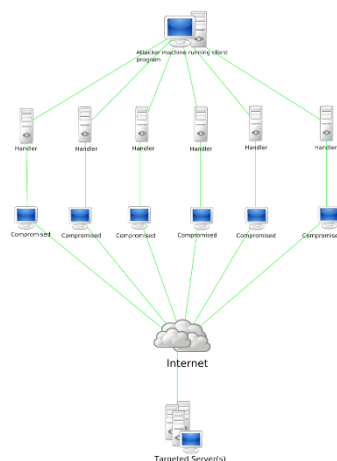
Furthermore, examples of what you have include physical security keys and other passkeys and examples of what you know include passwords or pins. However, to be effective, these protection methods need to be kept up to date, with certain organisational password policies forcing users to change their passwords every 14 or 30 days. In-person protection methods aim to increase security by preventing attackers from gaining physical access to an unauthorised area, such as door locks for highly vulnerable rooms such as server rooms, barbed wire and bollards to prevent entry into the building, and closed-circuit television (CCTV) cameras, which if placed in an area which you can easily see, can be used as a preventative measure but also as a measure to help identify attackers. Such protective measures are likely to also be legal requirements enforced by the government.

Phishing and social engineering attacks aim to gain information or access to data by manipulating individuals either in a company or personal setting. Companies provide training to improve individuals' awareness against certain types of attacks, such as not clicking unknown email attachments or sending confidential information over email to prevent phishing attacks, even if it seems like the request was sent from a trusted person. Further training could include teaching workers to not connect and use any devices like USB sticks, which they do not personally own as they could be infected with malware or intrusion software.

How can Artificial Intelligence be combined with Cybersecurity and Current Implementation?

This section aims to depict how the sectors of Artificial Intelligence and Cybersecurity can be combined to improve Cybersecurity techniques, including examples of how it has been used in current examples and analysis of how it will be used in the future as the importance of Cybersecurity becomes more paramount and the use of Artificial Intelligence within businesses increases.

A key area where Artificial Intelligence can be used is to improve protection against Distributed Denial of Service (DDOS) attacks. These attacks use a series of networks of devices which aim to disrupt the traffic of a website or server by flooding it with malicious traffic, causing the server to slow down or crash. This is shown in the figure (Coelho and Yellowlcon, 2008), which depicts an attacker machine sending commands to a series of compromised computers, which sends traffic across the internet to the targeted server.



An illustration of a DDOS attack (Coelho and Yellowlcon, 2008)

By analysing network traffic behaviour over time, Machine Learning algorithms learn what normal traffic is and identify deviations from that pattern. After identifying a suspected DDOS attack, Artificial Intelligence models can deal with the problem in real-time by blocking or rerouting the malicious traffic to ensure the effect of the attack is minimised. However, it is a significant challenge to differentiate between normal and malicious traffic, especially as they are likely to come from various devices from across the globe and do not seem linked and so look legitimate. This is especially evident as legitimate packets could be flagged and blocked during periods of high traffic as systems receive many similar requests from a large variety of devices and flood systems. This was particularly notable when cinema websites crashed from the high demand for tickets for the Avengers: Endgame movie in 2019, causing frustration as users struggled to buy tickets.

A key example of using Artificial Intelligence in Cybersecurity is the CrowdStrike Falcon (CrowdStrike,2024) endpoint and response detection system (ERD Systems), which includes Artificial Intelligence features but does not rely solely on these methods. ERD Systems like CrowdStrike Falcon work by continuously monitoring endpoints, any physical device which is connected and communicates with the network, checking for any threats and performing actions automatically, which can be performed using Artificial Intelligence programs, to reduce the risk of any of these vulnerabilities being exploited. For example, a snapshot of an the CrowdStrike Falcon dashboard indicates that the system had prevented 32 attacks in the last day and shows the user, typically a cybersecurity admin, the most recent detections for their review and generates a subjective “Crowdscore” to allow for comparison.

A key feature of ERD systems is the need to work constantly, so Artificial Intelligence can be used to reduce downtime in traditional systems. However, ERD Systems such as CrowdStrike Falcon may also have their own vulnerabilities, making them vulnerable to exploitation. For example, the CrowdStrike Falcon system has had issues that have caused global outages, particularly in July 2024, when a software configuration file caused errors that caused Microsoft Windows machines to crash and make them unusable. This had devastating effects as it grounded airline flights, disrupted healthcare services and American '911' emergency call centres, and took Sky News offline. This incident cost Fortune 500 companies over \$5 billion due to the disruptions (Ehrlich, C. 2024).

The computing giant Google also plans to implement Artificial Intelligence in its Cybersecurity measures using their Gemini generative Artificial Intelligence models. They aim to use semi-autonomous security to enable further Cybersecurity. A key feature of this is creating a playbook of response in the case of an attack which removes the need to hire workers to construct a playbook. This can then be simplified, and then its execution can then be simulated. This is expected to give security teams an improvement in the detection of attacks as well as improving their response.

There are also criminals who use key fears of Artificial Intelligence to enhance their attacks. As Natural Processing Languages enhance, enabling Artificial Intelligence models to understand and process texts and formulate a response, there is a fear that they can be used to enhance phishing attacks by reading and processing a hacked user's emails and being able to extract key information that can thusly be used for a more convincing attack. A key example of this is if a user has an email from a contractor or online marketplace asking for payment or a deposit for a good or service, and the Artificial Intelligence model formulates and sends an email to the victim to pay this deposit of the correct amount to a different account. This is likely more effective as it becomes harder to recognise that it is a fake email, especially as they ask for a

payment that the user is expecting, is for the correct amount and is for the correct good or service they intend to purchase. However, this has generated an argument for using Artificial Intelligence, which involves implementing counter-attack models to read emails and protect against such malicious emails created by Artificial Intelligence.

Artificial Intelligence models could also be incorporated into firewall systems, using Machine learning to train based on normal traffic and then simulations of attacks to see how well the Artificial Intelligence model identifies and protects against malicious traffic. In Anti-virus systems, Machine learning could also be used to train against aspects of known viruses to improve their recognition of similar viruses and reduce the false positives and negatives produced from traditional Anti-Malware programs.

Main Arguments for Artificial Intelligence in Cybersecurity

The main argument for using Artificial Intelligence is its necessity in combating AI-based cyberattacks. There is a critical concern that not incorporating AI models into Cybersecurity practices will, in turn, result in more significant vulnerabilities as attackers implement and improve their own AI-driven attacks.

For example, attackers have incorporated Artificial Intelligence models, particularly Large Language Models (LLMs) such as ChatGPT or Microsoft Copilot, in phishing attacks. These attacks have become more sophisticated through greater personalisation, leading to more convincing attacks. An example of this is where an LLM analyses and processes email conversations of a previous victim, allowing the model to build up a character profile of the victim, such as their specific language patterns, common phrases, and their email signature, allowing for a more convincing and targeted attack. Furthermore, they can also analyse the victim's relationships with their contacts, which becomes incredibly dangerous when impersonating individuals like contractors requesting payment for a job. The LLM creates a fraudulent email posing as the contractor, asking the victim for a deposit or payment for a job. The rate of social engineering attacks like these is likely to increase as the model can analyse and send out fraudulent emails at a faster rate than it would take a human to do so. This, as a result, advocates for the implementation of Artificial Intelligence-based fraudulent email detection systems which scan emails to detect whether an AI has written them and filter them as scams to prevent users from falling victim.

Additionally, Artificial Intelligence has been used to evolve Cybersecurity threat detection. This is evidenced by Buczak and Guven in 2016, who surveyed the use of Artificial Intelligence in Cybersecurity and concluded that Machine Learning is an effective method to analyse a large

amount of data in a short period of time as well as generating patterns of typical “green” traffic, that being traffic which is legitimate, and identify “red” traffic which is illegitimate or malicious or threats that traditional methods such as firewalls might miss. Firewalls may not work if attack traffic is funnelled around the firewall or where traffic is redirected from one port to another, allowing the attackers to bypass network security. The Machine learning models can be furthered as they learn by reinforcing their patterns as traffic is blocked in real time, increasing accuracy and reducing the number of false positives. A key use for Machine Learning Cybersecurity methods is in zero-day attacks – where attackers exploit a flaw before developers know about or can address the exploit whereas Machine Learning methods are more likely to be able to detect suspicious activity in those scenarios.

Shashank and Purnima (2019) support the use of Artificial Intelligence-driven endpoint protection platforms such as the CrowdStrike Falcon platform, which aims to protect users’ end devices such as laptops and phones. This is suggested to be done by continually monitoring device security as well as providing real-time threat detection and autonomous response, allowing networks that are particularly vulnerable to endpoint attacks to be constantly protected against malware and attacks. This has become increasingly important since the COVID-19 pandemic as more firms convert to being completely remote, meaning there is a greater challenge to protecting against malware as there has been a significant increase in the number of devices and networks which need to be protected and secured which could be located across the globe.

The accuracy of Artificial Intelligence models in Cybersecurity is likely to improve as more research is put into Natural Language Processing systems such as search assistants, email filters and data analysis to enhance the intelligence of an organisation’s security team. Mittal et al. (2016) describe Natural Language Processing as a branch of Artificial Intelligence which can be used to enhance threat detection by analysing unstructured data. Its implementation is

important as this allows unstructured and varied data, which is typical of network traffic, to be more accurately processed and analysed, allowing threat perception to become more accurate and allowing the model to learn from a wider range and type of media sources which enhances these models from being able to improve defences against several types of attacks like the aforementioned phishing attacks through email impersonations by processing emails and DDOS attacks by analysing network traffic further.

Counter Arguments of Artificial Intelligence in the Use of Cybersecurity Measures

As compared to a requirement to include Artificial Intelligence-based protection defences to defend against attacks, there is a risk of Adversarial Artificial Intelligence models which aim to attack Artificial Intelligence-based Cybersecurity methods. This is typically done with the use of machine learning models, which aim to exploit vulnerabilities in Artificial Intelligence defences by trial-and-error measures such as by providing flawed inputs into systems to gain unauthorised access to the system. These models are also furthered by their constantly learning nature, meaning their accuracy is likely to improve as they run.

Another significant counterargument of Artificial Intelligence is the fact that machine learning models require a large amount of accurate training data to be accurate and efficient. This creates a significant privacy concern as these models use web crawlers, a type of bot which browses and collects data from websites on the World Wide Web. This may lead to sensitive data being collected and processed by an individual or their corporation. This is especially significant as Artificial Intelligence models are also frequently targeted for attacks, meaning that if their training information is leaked, it could lead to sensitive information being leaked, which could be exploited.

Furthermore, these machine learning models can be deceived by adversaries into making incorrect assessments by providing inaccurate or misrepresentative training or learning data to deceive the models. A key example of this, as described by UC Berkley's Center for Long-Term Cybersecurity (2019), is Poisoning attacks, which incorrectly label data, causing the Artificial Intelligence model to make inaccurate decisions such as mislabelling threats as false positives. This can be hard to detect by human supervisors and can be especially vulnerable with unsupervised method training, a methodology where the model is trained without humans

adjusting their training data to ensure it is accurate. This allows the model to grow and improve at a significantly faster rate as they do not have to wait for a human supervisor to supply more training data; however, it makes the model more vulnerable to such poisoning attacks. A key example of this was the poisoning of the Microsoft 'Tay' chatbot in 2016, which was a Twitter chatbot which was programmed to learn to engage in conversation through interactions with other users. However, the chatbot lacked sufficient filters and so became flooded as thousands of people tweeted the account containing offensive or racist messages causing the chatbot's machine learning model to pick up this language and label it as typical language causing Tay's tweets to become increasingly offensive causing the chatbot to be shut down in less than a day.

There is also another risk of evasion attacks, which tend to take place after the machine learning system has been trained. This attack aims to sneak around the machine learning model's filters in a trial-and-error manner to bypass its filters, which could have the effect of a tunnelling attack, allowing attackers to get past the Cybersecurity measures in place.

An additional negative impact of Artificial Intelligence is that it is vulnerable to spoiling attacks. This means that for it to be effective, it is likely to have to be trained with selective training data and human supervisors. However, the efficacy of such a model depends on how easy it is for similar malware to be avoided. For example, if it is a simple match-up checker where the model checks whether the uploaded file matches a proven malware in its database, a key problem is whether the malware can easily be altered to pass through the malware check. There is also the additional issue that for such supervised methods to be effective, they need to be initially detected meaning they're likely to take a few weeks for them to join training data and therefore, the Artificial Intelligence model is likely to be playing 'catch up' to hackers, as well as proving no support against Zero Day attacks which haven't been discovered by security professionals before, generating a thought of whether the cost and time to implement such models are worth it at all.

Moreover, there is also a critical concern about the ethics and bias of Artificial Intelligence models. The Tony Blair Institute for global change published a report which estimates that Artificial Intelligence could displace between one and three million jobs over the next few decades, although occurring gradually (Sharps, 2024). As investment into such models increases, making them more accurate and cost-effective, there is likely to be greater unemployment in the Cybersecurity sector as firms aim to reduce their costs and lay off their more expensive human capital. This will result in a worsening of the lives of many as they see a fall in their disposable income, which becomes more significant in magnitude as the unemployment caused by Artificial Intelligence implementation is likely to be structural, a permanent decline in demand for their sector, which may mean that these workers are likely going to have to invest into new skills in order to find work. This can, in turn, slow down the economy as these workers produce less and consume less in the economy, as they do not have the funds to do so.

Overall Judgement and Conclusion

Considering both sides as to whether organisations should incorporate Artificial Intelligence models in their evolution of Cybersecurity methods has resulted in a conclusion which is in favour of the use of Artificial Intelligence models in Cybersecurity methodologies. The integration of Artificial Intelligence models, such as machine learning models, is likely to improve the accuracy of Cybersecurity efforts because of their consistent learning based on previous data. This is likely to improve the ability of security professionals to protect data from threats more quickly and efficiently as well as the implementation of autonomous protection methods which aim to eliminate threats in real time which reduces the effect that malicious cyberattacks can have.

For Artificial Intelligence integration in Cybersecurity to be effective, standards should be put in place to ensure safety and sufficient and accurate training data should be provided to avoid the model from being poisoned by malicious users. This sentiment is shared by Massachusetts Institute of Technology (MIT) professor and author Max Tegmark, who calls for funding for Artificial Intelligence safety research. He pursues this at the “Future of Life Institute,” which he founded himself. The institute aims to prevent the use of Artificial Intelligence from causing long-term extreme risks and using it to improve people's lives globally. This institute also published an open letter (Future of Life Institute, 2023) which called for governments to intervene and pause training of Artificial Intelligence models giving humanity enough time to decide that the benefits of these systems outweigh the negatives of these systems such as their ability to cause high levels of unemployment as they are more able to replace different jobs as a lower cost instead of promoting growth and development. To mitigate these negatives, there should be greater research into more diverse and inclusive data to reduce the effects of ethical bias caused by misrepresented training data. Furthermore, government organisations should be tasked with enforcing protocols to audit Artificial Intelligence systems, to identify any bias and keep businesses transparent. There should also be the promotion of human and Artificial

Intelligence collaboration, which could be achieved through a government subsidy to ensure Artificial Intelligence is used to help alongside humans and not cause significant unemployment spikes and to encourage the development of Artificial intelligence-related skills.

On the contrary, while it is essential for investments in greater research into Artificial Intelligence safety, malicious attackers are likely to implement Artificial Intelligence into their attacks at the earliest convenience, and as such, it is essential for organisations and security teams to implement countermeasures to malicious Artificial Intelligence models. However, one should not also discredit the ever-growing need for training within companies at every level because, in a scenario, it comes down to the awareness of workers to prevent an endpoint attack. Organisations should also not consider Artificial Intelligence to be the sole answer to their Cybersecurity problem, but rather only one part of the problem, with a greater need for investments into more traditional Cybersecurity practices.

Evaluation of my Dissertation

By writing this dissertation, the main skill I have been able to improve upon is academic research and referencing. My academic referencing skills were developed as I had to manage my references in a consistent manner and in Harvard style, which is something I would not typically have had to do normally in school studies. My academic research skills were developed by independent research to find relevant academic writings and articles and identify and interpret the relevant details of each source. When deciding which sources to use, I made sure to include a wide range of diverse sources. I ensured reliability by citing peer-reviewed and technical studies to balance the articles cited, which are not peer-reviewed and so can be less credible. However, geographical bias may have been introduced as many of my sources come from Western sources, particularly the United States and the United Kingdom. This may have resulted in an overly positive view of Artificial Intelligence as these are both very developed countries that are very eager to invest in it. My dissertation could have been improved by including sources from around the globe to include ideas from unfamiliar cultures and see how Artificial Intelligence systems may affect other countries. There may also have been bias caused by the citations used to define terms used as these were fundamental text sources, so my dissertation may have been improved by including more modern sources, so the definitions are more up to date.

A challenge that I faced while writing my presentation was presenting ideas and concerns from both arguments equally, as well as providing an overall judgement and insight into the future of the topic. This was due to the fact that Artificial Intelligence is a developing field that many firms and organisations are eager to implement, meaning it was difficult to find credible counterarguments about Artificial Intelligence as well as the fact that there are not a significant amount of credible real-world examples meaning that many counter-arguments are purely theoretical.

I found academic writing particularly difficult because it was challenging to write about different arguments I had encountered during my research and literature review without providing my own judgement and opinion.

To improve, I would like to have focused more on the future of Artificial Intelligence in Cybersecurity, such as case studies of its planned implementation in an organisational setting. As well as this, I feel that my dissertation could have been improved by using a wider range of types of academic documents as most of my references came from journal articles or news websites. If I had more time, it would have been useful to generate a framework of how Artificial Intelligence can be used in the future of Cybersecurity, including a description of examples of each stage of the framework as well as dictating how the advantages and challenges change over time. If I had more resources to improve my dissertation, I would have liked to research more into current methods that Artificial Intelligence is used in Cybersecurity and, conversely, in cyberattacks. I would have been able to develop a more thorough dissertation with personal experience of different types of cyberattacks and Artificial Intelligence-based protection methods, where I would have been able to test their efficacy against different attacks on a private network in a virtual sandbox.

Definitions

Cybersecurity - Methods used to protect and secure systems against cyber-attacks.

Cyberattack - An attack by criminals to gain unauthorised access to a system, compromising its confidentiality, integrity, or availability.

CIA Triad – Three aspects that every Cybersecurity measure aims to attain and protect:

Confidentiality, Integrity, and Availability

Confidentiality – Preventing unauthorised users from accessing a system, area, or data.

Integrity—Ensuring that data is complete and has not been altered by an unauthorised user and making the authorised user responsible for that data.

Availability – Making sure data can be accessed by an authorised user whenever they would like to access it.

Cybercriminals – Someone who uses computers or the internet to commit malicious activities.

Anti-virus software – Programs which aim to detect and quarantine computer viruses.

Artificial Intelligence – Computer systems which can perform tasks which normally require human intelligence.

False positive – Where a security system incorrectly flags a legitimate file as malicious, often used to quantify the efficacy of certain Cybersecurity methods.

Firewall—Cybersecurity software that monitors network traffic based on set security rules so only legitimate traffic passes through to the internal network.

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